

Finite Element Methods – APE3

Hilbert and Sobolev Spaces

Objectives of the session:

- Use the course material to prove existence of solutions via the Lax–Milgram theorem.
- Manipulate the concepts related to Hilbert spaces.
- Verify the assumptions of the Lax–Milgram theorem using results from the course or known algebraic results.
- Get a better overview of the Poincaré inequality

There exists a constant $C_P > 0$ (depending only on the dimension and Ω) such that

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_0^1(\Omega).$$

1 Let us solve linear systems using Lax–Milgram

Exercise idea. Solve the linear system

$$Ku = f, \quad K \in \mathbb{R}^{n \times n}, \quad f \in \mathbb{R}^n, \quad u \in \mathbb{R}^n,$$

using a very (too!) powerful tool: the Lax–Milgram theorem. Clearly, the system admits a solution if and only if $\det K \neq 0$.

1. Recall why $\mathbb{R}^n$, equipped with the scalar product $(u | v)$, is a Hilbert space.
2. Define

$$a(u, v) = (Ku | v), \quad \ell(v) = (f | v),$$

and show that if a and ℓ satisfy the assumptions of the Lax–Milgram theorem, then there exists a unique $v \in \mathbb{R}^n$ solving the problem $Kv = f$.

3. (a) Show that $\ell(v)$ is bounded in the sense of the theorem. Which inequality do you use?
solve
- (b) Show that $a(u, v)$ is continuous. Which inequality do you use?
(You will need $\|Ku\| \leq \|K\|\|u\|$. What is the meaning of $\|K\|$?)
- (c) Assume that K is symmetric. Using the diagonalization theorem, show that K is coercive.
How does the coercivity constant β depend on the eigenvalues?
(You may use that $Q^T Q = \text{Id}$ for an orthogonal matrix Q .)
- (d) Let K be a general square matrix. Define

$$K_{\text{sym}} = \frac{K^T + K}{2}.$$

Show that

$$a(u, u) = (Ku \mid u)$$

depends only on K_{sym} .

4. State a condition on the eigenvalues of K_{sym} ensuring that $a(\cdot, \cdot)$ is coercive.
5. Is this condition equivalent to the nonvanishing of $\det K$, the product of the eigenvalues?

2 Let us solve $-\Delta u = f$ using Lax–Milgram

Exercise idea. Find a solution (*i.e.* show existence) to the boundary value problem below using a powerful functional analytic tool: the Lax–Milgram theorem. The idea is to rewrite the equation in weak form and interpret it as a Lax–Milgram problem on the Hilbert space $H_0^1(\Omega)$.

Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain. We consider the problem ($f \in L^2(\Omega)$)

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The associated weak formulation is: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) := \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx := \ell(v) \quad \forall v \in H_0^1(\Omega).$$

1. Recall the two definitions of $H_0^1(\Omega)$ and recall the inner product of this Hilbert space.

Explain why this space properly encodes the homogeneous Dirichlet boundary condition. Discuss why the weak formulation is elegant: the boundary condition is built into the functional framework, so it cannot be forgotten.

2. Define

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \ell(v) = \int_{\Omega} f v \, dx,$$

and show that if a and ℓ satisfy the assumptions of the Lax–Milgram theorem, then there exists a unique $u \in H_0^1(\Omega)$ solving the weak problem.

Does this weak solution solve the PDE in the classical sense?

3. Show that $\ell(v)$ is bounded in the sense of the theorem. Which inequalities do you use?
4. Show that $a(u, v)$ is continuous. Which inequality do you use?
5. Show that $a(u, v)$ is coercive. Which inequality do you use?
6. Let V be a finite-dimensional subspace of $H_0^1(\Omega)$; this is the idea behind the Galerkin method and the finite element method. Does there exist a unique solution $u \in V$ to

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in V?$$

3 Let us solve $-\Delta u + Vu = f$ using Lax–Milgram

Exercise idea. Show existence to the system below by *perturbing* the arguments from the previous section.

Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain, and let $V \in C^0(\bar{\Omega})$ be a given potential. We consider the problem ($f \in L^2(\Omega)$)

$$\begin{cases} -\Delta u + Vu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The associated weak formulation is: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) := \int_{\Omega} \nabla u \cdot \nabla v \, dx + \int_{\Omega} Vuv \, dx = \int_{\Omega} fv \, dx = \ell(v) \quad \forall v \in H_0^1(\Omega).$$

1. Recall that $\ell(v)$ is bounded in the sense of the theorem by using the result of the previous section.
2. Show that $a(u, v)$ is continuous.
3. Show that if $V \geq 0$ for all $u \in H_0^1(\Omega)$, $a(u, u)$ is coercive.
4. Show that for all $u \in H_0^1(\Omega)$, $a(u, u)$ satisfies $a(u, u) \geq (1 - C_P^2 \|V_-\|_{\infty}) \|\nabla u\|_{L^2(\Omega)}^2$, and state a condition on V for a to be coercive. *The condition should be weaker than $V \geq 0$.*
5. Using your condition on V state that there exists a unique solution to the weak form of the problem using the Lax-Milgram theorem.